



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

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SPECIFICATION

FOR

PERSONAL PROTECTIVE EQUIPMENT

FALL PREVENTION FROM A HEIGHT

(HARNESSES)

Issue: Jan.-2023/ Rev- 1

- This revision contains option items that must be selected by EDC before bidding.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

CONTENTS

CLAUSE	DESCRIPTION	PAGE
1	SCOPE	3
2	APPLICABLE STANDARDS	3
3	ENVIRONMENTAL CONDITIONS	3
4	CONSTRUCTION REQUIREMENTS	4
5	PERFORMANCE REQUIREMENTS	7
6	MARKING	7
7	TESTING and INSPECTION	8
8	SUBMITTALS	9
9	GUARANTEE	10
10	PACKING and DELIVERY	10
11	TECHNICAL DATA SCHEDULE	10



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

1. SCOPE

This specification covers the minimum technical requirements for design, manufacturing, testing, and supply of full body harnesses as personal protective equipment (PPE). The required harnesses are intended for the purpose of work positioning and preventing workmen of Egyptian Electricity Distribution Companies (EDCs) against the hazard of a fall from a height.

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
BS EN 358	Personal protective equipment for work positioning and prevention of falls from a height – Belts and lanyards for work positioning or restraint
BS EN 361	Personal protective equipment for work positioning and prevention of falls from a height – Full body harnesses
BS EN 362	Personal protective equipment for work positioning and prevention of falls from a height – Connectors
BS EN 354	Personal fall protection equipment — Lanyards
BS EN 355	Personal protective equipment for work positioning and prevention of falls from a height – Energy absorbers
BS EN 364	Personal protective equipment for work positioning and prevention of falls from a height – Test methods

3. ENVIRONMENTAL CONDITIONS

The safety behavior of the required harness shall be guaranteed under the environmental conditions given in Table (2).



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

Table (2)

Ambient temperature	-5°C to +45°C (50°C is an option according to __ EDC requirements), (35°C monthly average of the hottest month, 25°C yearly average)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level
Environment	Humid tropical climate with heavily polluted atmosphere

4. CONSTRUCTION REQUIREMENTS

- Full Body Harness shall consist of the harness and a fall protection lanyard (with an energy absorber). According to __ EDC requirements, full body harness (including fall protection lanyard) can be equipped with back support and a work positioning lanyard (refer to Figure 1).

4.1 Full Body Harness

- A full body harness shall comprise straps or similar elements which are placed in the pelvic area and on the shoulders. The full body harness shall fit the wearer. Means of adjustment may be provided.
- Straps shall not migrate from position and shall not loosen by themselves.
- The width of primary straps shall be at least 40 mm and of secondary straps at least 20 mm.
- Webbing and yarns shall be made from virgin filament or multifilament synthetic fibers suitable for the use intended. The breaking tenacity of the synthetic fiber shall be known to be at least 0.6 N/tex.
- Threads used for sewing shall be physically compatible with the webbing and of a quality comparable with that of the webbing. They shall be of a contrasting color or shade in order to facilitate visual inspection.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

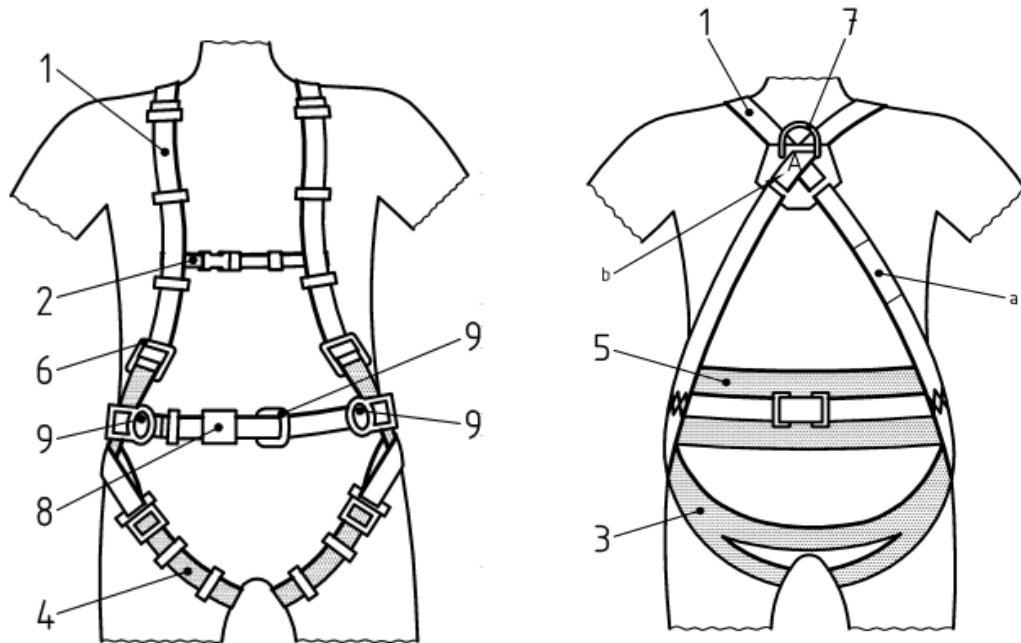


Figure (1) Example of a full body harness with back attachment for fall arrest and attachment for work positioning

Key:

- | | |
|---|----------------------|
| 1 Shoulder strap | 2 Secondary strap |
| 3 Sit strap (Primary strap) | 4 Thigh strap |
| 5 Back support for work positioning | 6 Adjustment element |
| 7 Fall arrest attachment element | 8 Buckle |
| 9 Attachment element for work positioning | |
| a and b Markings (General and specific information as per clause 6) | |

4.2 Lanyards

- Lanyards made of textile materials (e.g. synthetic fiber) shall be able to sustain a force of 22 kN when tested as per EN 354 or similar test method. In addition, Lanyards shall be able to retain a mass of 100 kg when dropped from a height of 4 m when tested as per EN 354 or similar test method.
- The lanyards described above shall have a maximum length of 2 m.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

4.3 Connectors

- Connectors shall not have sharp edges or burrs that may cause injury to the user or that may cut, abrade or otherwise damage webbing or rope. Materials, which may come into contact with the skin of a user, shall not be known to cause irritating and sensitization effects during normal use of the connector
- Gates in the connectors shall be self-closing (class T as per EN 362) and shall have an automatic self-locking feature (refer to Figure 2). Self-locking feature shall lock the gate automatically when the gate shuts and shall require at least two different deliberate manual actions to open the gate.
- Screw link connectors and manual-locking gates are not accepted.

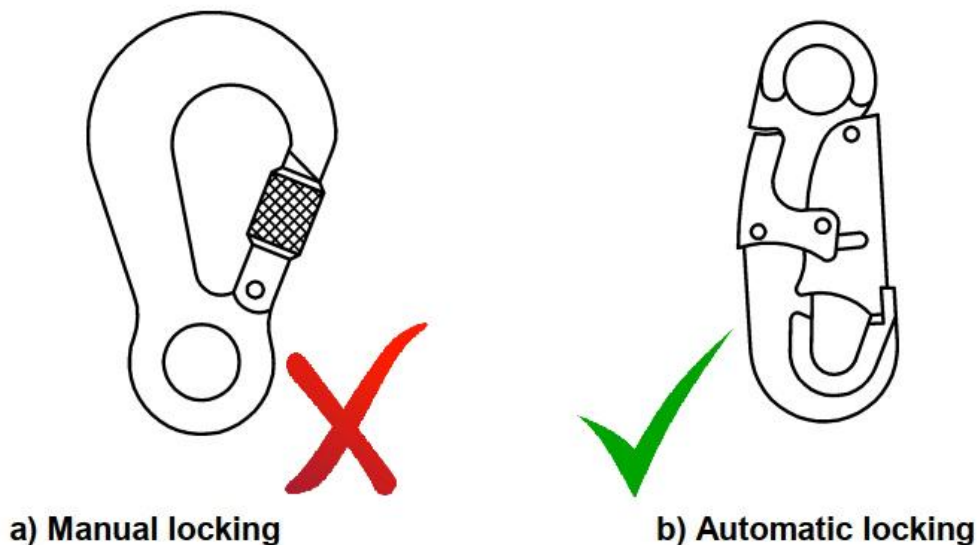


Figure (2) Screw link (manual-locking gates) vs (Automatic locking gates)

4.4 Energy Absorbers

- The full body harness should be provided with an energy absorber. The energy absorber should be integral with the fall protection lanyard. (i.e. the energy absorber cannot be removed without mutilating the lanyard).



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

5. **PERFORMANCE REQUIREMENTS**

- a) **Static strength test:** Full body harness shall not release a dummy torso (of 100 kg) under a force of 15 kN as per EN 364 or similar test method.
- b) **Dynamic Strength Test:** Full body harness shall withstand two successive drop tests of a dummy torso (of 100 kg) when dropped from 4 m height as per EN 364 or similar test method.

6. **MARKING**

- Each item of PPE or other equipment shall be clearly, indelibly and permanently marked by the manufacturer and shall include the following general information in addition to specific information:

6.1 **General Information**

- a) Manufacturer's name, PPE type and model
- b) Standard (include document date) to which the equipment conforms
- c) Indication to the necessity for users to read the instructions for use.
- d) Month and year of manufacture

6.2 **Specific Information**

- **Full Body Harness:** A capital letter "A" at each fall arrest attachment element
- **Connectors:** If the minimum strength claimed by the manufacturer for the major axis is marked on the connector, the marking shall be for the closed and locked position (refer to Figure 3). The marked strength shall be in whole number kN.

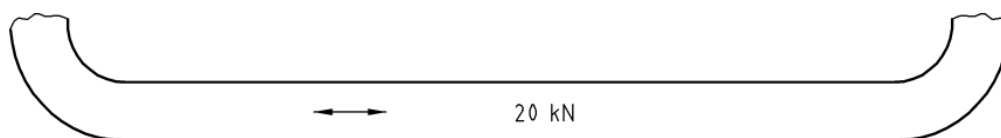


Figure (3) Example of marking of major axis strength with gate closed and locked



EEHC DISTRIBUTION MATERIALS SPECIFICATION

**EDMS 30-302-1
10-01-2023**

- Lanyard: Maximum lanyard length
- Energy Absorber: Maximum length allowed of the energy absorber including lanyard.

7. TESTING and INSPECTION

7.1 General

- Full Body Harness shall be subjected to the following tests in accordance with the latest relevant standard and as specified herein.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Tests shall be carried out by an accredited independent testing laboratory (or other testing laboratories accepted by __ EDC).
- Tests shall be carried out in the presence and under the control of the inspecting engineer of __ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by __ EDC.
- The contractor should advise testing dates. The contractor shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- Test reports shall clearly show the equipment concerned, the manufacturer's identity, the tests carried out, test results and the standard's requirements against the test results to determine passing or failing of the test.

7.2 Tests

- The following tests shall be carried out for full body harness in accordance to EN Standard:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

- a. Static Strength of Full Body Harness
- b. Dynamic Strength of Full Body Harness
- c. Static Strength of Lanyard
- d. Dynamic Strength of Lanyard
- e. Slippage for Lanyards with Length Adjustment Device
- f. Corrosion Resistance of Metal Parts
- g. Static Strength of Connectors
- h. Gate Function of Connectors
- i. Corrosion Resistance of Connectors
- j. Static Preloading of Energy Absorbers
- k. Dynamic Performance of Energy Absorbers
- 1. Static Strength of Energy Absorbers

8. SUBMITTALS

8.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data and original catalogues shall be submitted with the offer.
- **One specimen** of the offered equipment (including all accessories) shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of test reports.
- List of deviations & clauses to which exception is taken (if any).



EEHC DISTRIBUTION MATERIALS SPECIFICATION

**EDMS 30-302-1
10-01-2023**

8.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Packing details.
- The supplier should provide all the instructions of use/storing in (Arabic and English).

9. GUARANTEE

- The supplier guarantee the supplied materials and all accessories against any defects arising out of faulty design or workmanship, or of defective material for a period of one (1) year at least from the date of delivery.

10. PACKING and DELIVERY

- Full body Harness shall be packaged in an individual container or package of sufficient strength to properly protect the product from damage. The outside of the container or package shall be marked as specified in relevant standard.
- The type of packaging suitable for transport shall be defined by the manufacturer.
- At the request of the customer or according to government specifications any additional or amended instructions shall be included in the package.

11. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

TECHNICAL DATA SCHEDULE
PERSONAL PROTECTIVE EQUIPEMENT (PPE)

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	__ EDC Requirement*	Vendor Offer**
1	Required PPE Item	○ Full Body Harness (including Fall Protection Lanyard) ○ Full Body Harness (including Fall Protection Lanyard) together with back support for work positioning (including Work Positioning Lanyard) ○ Energy Absorber for Fall Protection Lanyard	
2	Manufacturer's name		
3	Model		
4	Country of Origin		
5	Applicable Standard	○ BS EN 361 - Full body harnesses ○ BS EN 362 – Connectors ○ BS EN 354 – Lanyards ○ BS EN 355 Energy Absorber	
6	Full Body Harness		
6.1	Width of the primary straps	minimum 40 mm	
6.2	Width of the secondary straps	minimum 20 mm	
6.3	Breaking force of the material of the Lanyard	minimum 22 kN	
6.4	Maximum length of the lanyard	2 m	



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-302-1
10-01-2023

7	Connectors		
7.1	Gate closing and locking	Self-closing and self-locking	
7.2	Gate opening	Require at least two different deliberate manual actions to open the gate	
8	Year and Month/Quarter of Manufacture	Manufactured within two years before the bid closing date.	
9	Marking on the equipment	Please refer to clause 6	
10	Test Reports	Please refer to clause 7	
11	Specimen	Please refer to clause 8	
12	Guarantee	Please refer to clause 9	
13	Packing and delivery	Please refer to clause 10	
14	Submittals	Please refer to clause 8	

(*) - Values to be provided / proposed by the vendor.

(**) – Please, provide explanations for deviations if any. (Use separate sheet if necessary.)

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		